SILICON PHOTONICS

Project

Analysis of a Hybrid VO2/Silicon 2x2 Microring Switch

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EQUATIONS

Through port (desired output)

$$P_{out} = a(1 - t^2)^2 / 1 - 2t^2 a\cos(\phi) + (t^2 a)^2 \quad \text{---------- (1)}$$

Drop port (crosstalk / leakage)

$$P_{cross} = t^2 a^2 - 2t^2 a\cos(\phi) - t^2 / 1 - 2t^2 a\cos(\phi) + (t^2 a)^2 \quad \text{---------- (2)}$$

Loss conversion

$$a = 10^{-A(dB)/20} \quad \text{---------- (3)}$$

CROSS STATE:

$$P_{out,cross} = a_i(1 - t^2)^2 / 1 - 2t^2 a_i \cos(\phi) + (t^2 a_i)^2 \quad \text{---------- (4)}$$

$$P_{crosstalk,\,cross} = t^2 a_i^2 - 2t^2 a_i \cos(\phi) - t^2 / 1 - 2t^2 a_i \cos(\phi) + (t^2 a_i)^2 \quad \text{---------- (5)}$$

$$a_i = 10^{-A_i/20} \quad \text{---------- (6)}$$

BAR STATE:

$$P_{crosstalk,\,bar} = a_m(1 - t^2)^2 / 1 - 2t^2 a_m \cos(\phi) + (t^2 a_m)^2 \quad \text{---------- (7)}$$

$$P_{out,\,bar} = t^2 a_m^2 - 2t^2 a_m \cos(\phi) - t^2 / 1 - 2t^2 a_m \cos(\phi) + (t^2 a_m)^2 \quad \text{---------- (8)}$$

$$a_m = 10^{-A_m/20} \quad \text{---------- (9)}$$

Insertion loss:

$$IL = 10\log_{10}(P_{desired}) \quad \text{---------- (10)}$$

Crosstalk:

$$XT = 10\log_{10}(P_{leak}) \quad \text{---------- (11)}$$

INSERTION LOSS - CROSS STATE:

$$IL = 10\log_{10}(P_{out,\,cross}) \quad \text{---------- (12)}$$

$$IL = 10\log_{10}(P_{XT,\,cross}) \quad \text{---------- (13)}$$

INSERTION LOSS - BAR STATE:

$$IL = 10\log_{10}(P_{out,\,bar}) \quad \text{---------- (14)}$$

$$IL = 10\log_{10}(P_{XT,\,bar}) \quad \text{---------- (15)}$$

INTRODUCTION

Silicon photonic switches are fundamental components in modern optical communication and signal processing systems, enabling high-speed data routing, wavelength multiplexing, and on-chip optical interconnects. Silicon-on-insulator (SOI) technology offers strong optical confinement, compact device footprints, and full compatibility with CMOS fabrication processes, making it a leading platform for large-scale photonic integration. Among the various switching architectures, microring resonator–based switches are particularly attractive due to their wavelength selectivity, small footprint, and low insertion loss.

Conventional silicon microring switches typically rely on thermo-optic or carrier-injection mechanisms to achieve reconfigurability. While effective, these approaches suffer from inherent limitations, including relatively high power consumption, slow switching speeds, and thermal crosstalk, which restrict their suitability for next-generation high-throughput photonic systems. These challenges have motivated the exploration of alternative tuning mechanisms and novel material platforms that can enable faster, more energy-efficient switching.

Vanadium dioxide (VO_2) has emerged as a promising phase-change material for photonic applications due to its abrupt and reversible insulator–metal transition, which induces large variations in both optical absorption and refractive index over a small temperature or electrical bias range. When integrated with silicon photonic devices, VO_2 enables switching dominated by loss modulation rather than phase tuning, allowing extremely short active lengths and ultra-compact device designs. Hybrid VO_2/silicon microring resonators exploit this property to achieve high extinction ratios, low power consumption, and reduced device footprints compared to purely silicon-based solutions.

In particular, the hybrid VO_2/silicon 2×2 microring switch represents an efficient architecture for compact optical routing. By selectively modulating the optical loss of the microring through the VO_2 phase transition, the device can toggle between bar and cross states with minimal structural complexity. The analytical framework presented in *"Analysis and Design Optimization of a Hybrid VO_2/Silicon 2×2 Microring Switch"* (April 2016) provides closed-form expressions for the output power, insertion loss, and crosstalk, enabling systematic exploration of design trade-offs without relying on computationally intensive full-wave electromagnetic simulations.

In this work, an analytical and numerical investigation of the hybrid VO_2/silicon 2×2 microring switch is carried out using MATLAB. The governing equations describing the switch operation are implemented and used to evaluate insertion loss and crosstalk under resonance conditions. Key design parameters, including the transmission coefficient of the directional coupler, the loss contrast between VO_2 insulating and metallic states, and the role of phase variation, are systematically analyzed. In addition, the influence of microring radius on thermal switching speed is examined to verify the motivation for compact device geometries, showing that smaller radii enable faster thermal response due to reduced active volume. By reproducing and extending the trends reported in the reference work through physics-based modeling, this study demonstrates the effectiveness of MATLAB-based analytical tools for early-stage design optimization and performance assessment of hybrid VO_2/silicon photonic switches.

WORKING PRINCIPLE

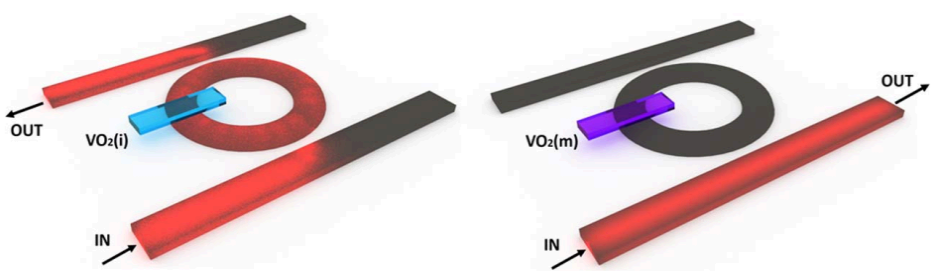
The hybrid VO_2/silicon 2×2 microring switch consists of a silicon microring resonator side-coupled to two straight bus waveguides, forming a four-port switching structure. Light entering the input port can be routed to either the bar or cross output port depending on the optical state of the microring. The switching operation is controlled by a short VO_2 section integrated along the microring waveguide, whose optical properties change due to the insulator–metal transition.

The directional couplers between the bus waveguides and the microring are characterized by a transmission coefficient t and a coupling coefficient k, satisfying $t^2 + k^2 = 1$. The microring introduces a round-trip amplitude attenuation a, accounting for propagation losses and VO_2 absorption, and a round-trip phase shift ϕ, determined by the effective refractive index and operating wavelength. The amplitude loss factor is expressed as in equation-3, where A_{dB} is the round-trip loss in decibels.

Using standard microring resonator theory, the normalized output powers at the bar and cross ports are described by closed-form expressions derived in the reference paper. These expressions relate the output power to the coupling coefficient t, loss factor a, and phase shift ϕ, and form the basis of the analytical model implemented in MATLAB. The switch performance is evaluated in terms of insertion loss (IL) and crosstalk (XT), defined as in equations-(10) & (11), where $P_{desired}$ is the power at the intended output port and P_{leak} is the unwanted leakage power at the opposite port.

Switching is achieved by altering the optical loss and phase of the microring through the VO_2 phase transition. In the cross state, VO_2 remains in its insulating phase, resulting in low absorption and minimal phase perturbation. Under resonance conditions (ϕi=0), the microring efficiently couples light to the cross port. In the bar state, VO_2 transitions to the metallic phase, introducing a large increase in optical loss and a moderate phase shift. This suppresses the resonant buildup in the ring, allowing most of the input light to pass directly to the bar port.

The dominant switching mechanism in this device is loss modulation rather than phase tuning. However, phase variations introduced by VO_2 can influence the resonance condition and therefore affect insertion loss and crosstalk. To capture this behavior, both loss-only switching and combined loss–phase switching scenarios are evaluated analytically by varying the loss contrast and phase difference between the two VO_2 states. This analytical framework enables systematic exploration of design trade-offs and forms the foundation for the parametric studies presented in the subsequent sections.

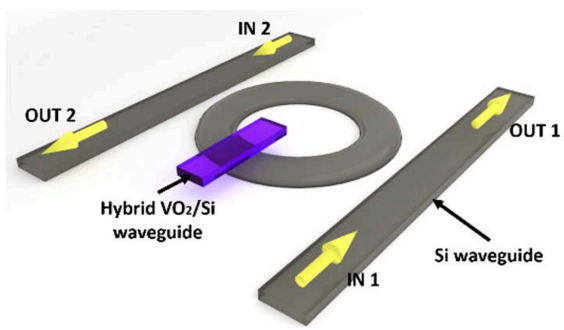


Switch operation in the cross state ($VO_2(i)$) and the bar state ($VO_2(m)$) -*(Ref.[1])*

Role of Coupling and Loss Balance:

The performance of the hybrid VO_2/silicon microring switch is governed by three key parameters: the transmission coefficient t of the coupler, the round-trip amplitude loss of the ring, and the phase shift ϕ. The transmission coefficient controls the coupling strength between the bus waveguide and the ring, determining how efficiently light can build up inside the resonator. In the cross state, VO_2 is in the insulating phase, resulting in low loss and an approximately zero phase shift, which supports strong resonance and efficient power transfer to the cross port. In the bar state, VO_2 becomes metallic, introducing significantly higher optical loss and a finite phase shift, thereby suppressing resonance and favoring direct transmission to the bar port.

Using standard microring resonator equations, the desired output power and leakage power are calculated, from which insertion loss and crosstalk are defined as $-10\,log_{10}$ of the respective powers. By sweeping the transmission coefficient t, a balanced operating point is identified where the insertion losses of the bar and cross states become equal. This occurs because weaker coupling compensates for the increased loss in the metallic state while maintaining resonance in the insulating state. For the TE-polarized design, this condition is achieved at approximately t≈0.88t, as confirmed by the MATLAB simulations. Additionally, the phase shift introduced by VO_2 in the metallic state further detunes the resonance, reducing unwanted coupling and improving crosstalk suppression, thereby enhancing overall switching performance.



Reference from [1]

SIMULATION - PLOTS

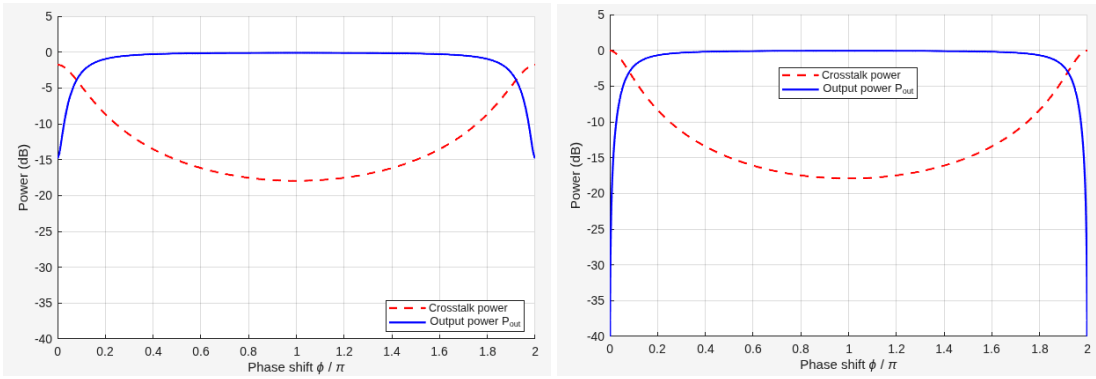


Fig.1: Normalized output power and crosstalk for a value of Ai = 20 $log_{10}(a_i)$ = 0.5 & 0.01 dB and a transmission coefficient (t) of 0.88

While placing the input signal exactly at resonance (φ = 0) maximizes power buildup in the ring and gives very low insertion loss, the bar-port crosstalk remains relatively high. By slightly detuning the input (φ = 1), the output at the cross port remains high (0 dB), but crosstalk drops significantly (−18 dB), providing better isolation. This illustrates the **trade-off between insertion loss and crosstalk**, and why practical devices may operate slightly off-resonance for optimal switching performance.

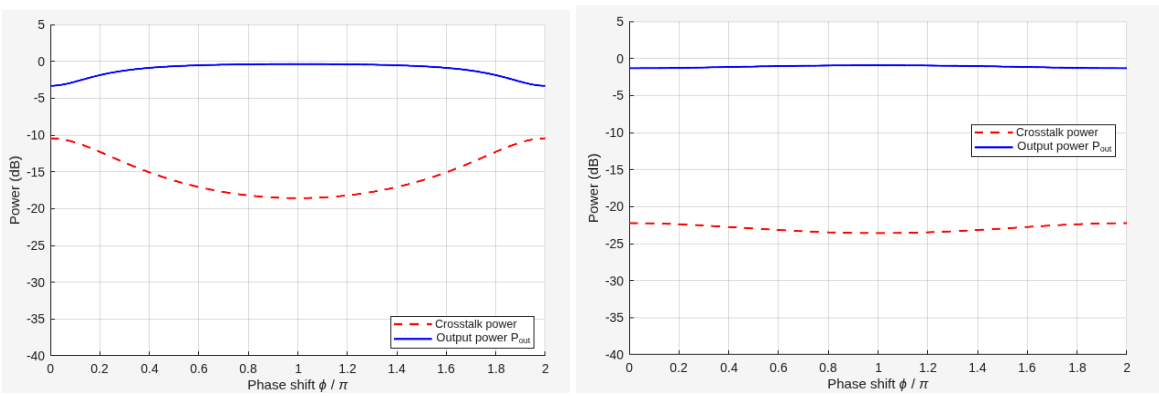


Fig.2: Switching performance for a value of Am = 20 $log_{10}(a_m)$ = 5 & 20 dB and the same transmission coefficient as in Fig. 1.

For a metallic-state VO_2 loss of 5 dB and the same coupler design, the real part of VO_2's refractive index changes, introducing a phase shift compared to the insulating state. This detunes the ring resonance, which **reduces crosstalk and improves insertion losses**. By carefully designing the directional coupler transmission, it is possible to balance insertion losses for both switching states, achieving symmetric and low-loss switching performance.

The parameters Ai and Am represent the **round-trip optical losses of the microring resonator** when VO_2 is in the insulating and metallic states, respectively. They are defined as $A = 20\,log_{10}(a)$, where a is the round-trip **amplitude transmission factor** of the optical field. Physically, these losses account for the total attenuation accumulated over one circulation of the ring, including silicon waveguide propagation loss, bending loss, scattering from fabrication imperfections, and absorption introduced by the VO_2 layer. In the insulating state, VO_2 exhibits a relatively small imaginary refractive index, resulting in weak absorption. Consequently, a low round-trip loss of Ai=0.5 dB is assumed, which is consistent with values reported in the literature for silicon microrings integrated with thin VO_2 films and represents a realistic low-loss operating condition.

When VO_2 transitions to the metallic state, its imaginary refractive index increases significantly, leading to much higher optical absorption within the ring. This increased attenuation is captured by the metallic-state loss parameter Am, which is set to Am=5 dB in accordance with the reference paper. The large loss contrast $\Delta A = A_m - A_i$ is the primary mechanism responsible for switching between the cross and bar states, as it suppresses resonance in the metallic phase and forces the optical power to remain in the bar port. While Ai fundamentally limits the minimum achievable insertion loss in the cross state, Am determines the extinction ratio and crosstalk performance in the bar state. Therefore, the selection of realistic values for both Ai and Am is essential to accurately capture the loss-driven switching behavior and coupling-loss trade-offs in hybrid VO_2/silicon microring switches.

Resonance plays a central role in the operation of the VO_2/silicon microring switch. In the cross state, when VO_2 is in the insulating phase, the round-trip loss is low and the phase condition for resonance is satisfied (ϕ_i **≈ 0**), allowing strong optical field buildup inside the ring. This resonant enhancement enables efficient power transfer from the input waveguide to the cross port, resulting in low insertion loss and high crosstalk suppression at the bar port. When VO_2 transitions to the metallic state, two effects occur simultaneously: the increased absorption raises the round-trip loss, suppressing the resonance, and the change in the real part of the refractive index introduces an additional phase shift (ϕ_m **>** ϕ_i), which detunes the resonance from the operating wavelength. This detuning further reduces the circulating power in the ring, causing most of the optical power to remain in the bar port. Although loss modulation alone is sufficient to achieve switching, resonance detuning enhances isolation and crosstalk suppression, particularly for moderate loss contrasts, thereby improving the overall switching performance.

State	**VO2**	**Ring loss**	**Phase Shift**	**Effect on power**
Cross	Insulating	$A_i = 0.5\,dB$	$\phi_i = 0$	Max power transfer to cross port, minimal insertion loss, higher crosstalk
Bar	Metallic	$A_m = 5\,dB$	$\phi_m > \phi_i$	Resonance detuned → lower cross-port power, lower bar-port leakage (better crosstalk)

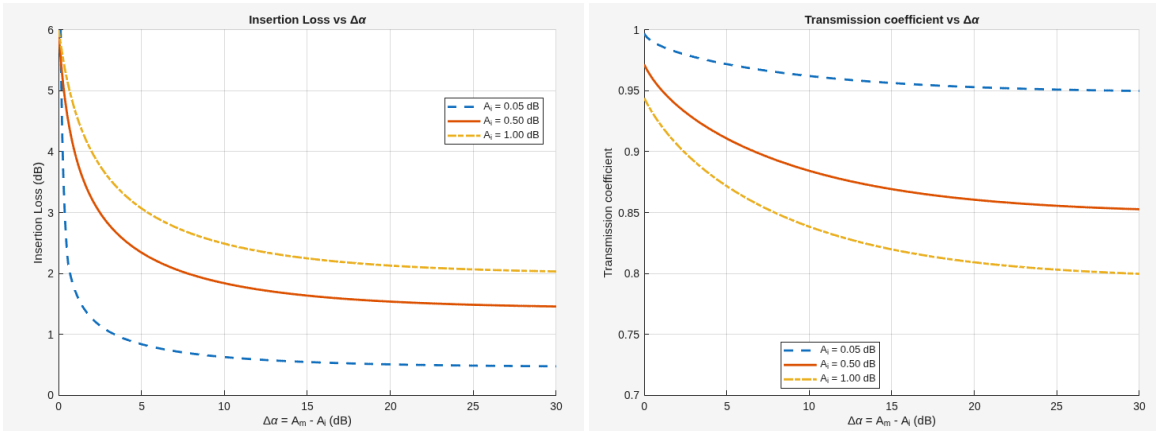


Fig.3: Insertion losses and transmission coefficient as a function of the ring loss difference ($\Delta\alpha$) between the bar and cross states by assuming that the phase-shift variation is zero ($\Delta\Phi$ = 0).

($\Delta\alpha = A_m - A_i$ ***,*** $\Delta\Phi = \Phi_m - \Phi_i$ ***, Critical coupling (Δα = 0): t =*** $\sqrt{a_i}$ ***)***

Figure 3 illustrates the dependence of insertion loss and transmission on the ring loss difference $\Delta\alpha = A_m - A_i$ under the condition of zero phase shift (Δϕ = 0), thereby isolating the effect of **pure loss modulation** on the switching behavior. The results clearly demonstrate that switching between the bar and cross states can be achieved solely by varying the optical absorption in the ring, without relying on resonance detuning. As Δα increases, the resonance in the metallic state becomes progressively more suppressed, forcing optical power to remain in the bar port while preserving resonant transfer in the insulating state. This leads to simultaneously lower insertion loss in the cross state and higher crosstalk suppression in the bar state, confirming that absorption contrast is the dominant switching mechanism in hybrid VO_2/silicon microring switches.

A key insight from Fig. 3 is that the **minimum achievable insertion loss is fundamentally limited by the insulating-state loss Ai**. Even with very large Δα, insertion losses cannot be reduced below the value imposed by Ai, because the cross-state performance depends on how efficiently the ring can sustain resonance. This figure effectively consolidates the physics shown separately in Figs. 1 and 2 by expressing the switching behavior in terms of a single parameter, Δα, which captures the combined influence of the insulating and metallic states. While the transmission coefficient **t** determines how losses are balanced between the two states, Fig. 3 highlights that optimizing the intrinsic material losses Ai and Am is a prerequisite for low-loss, high-contrast switching, independent of the specific coupling design.

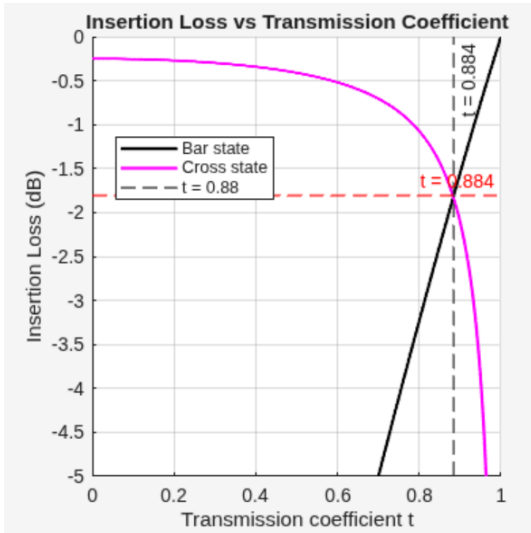


Fig.4: Insertion Loss vs. Transmission Coefficient in VO2-Si Microring Switch (Cross and Bar States)

Figure 4 illustrates insertion loss versus transmission coefficient *t* (swept 0.75 - 0.98) for the hybrid VO2-Si microring switch in cross and bar states. Cross state features VO2 insulating with low loss *Ai*=0.5 dB and phase *ϕi*=0, promoting efficient ring coupling and cross-port output. Bar state employs VO2 metallic with high loss *Am*=5 dB and phase *ϕm*=0.2, suppressing resonance for bar-port transmission (*a*=10^−*A*/20). Curves intersect at optimal *t*≈0.88, equalizing losses (~1-2 dB) via microring theory balancing coupler self-coupling against VO2-modulated absorption and phase.

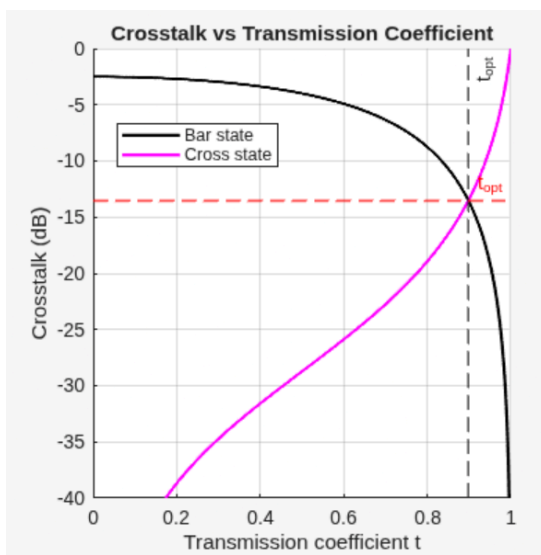


Fig.5: Crosstalk vs. Transmission Coefficient in VO2-Si Microring Switch (Cross and Bar States)

Figure 5 , the complementary plot to Figure 5 depicts crosstalk versus *t* (0.75-0.98) across states, using same parameters: cross (*Ai*=0.5 dB, *ϕi*=0) and bar (*Am*=5 dB, *ϕm*=0.2). Crosstalk arises from leakage power, with port roles swapped in bar state per standard resonator formulas written at the beginning. At *t*≈0.88, insertion losses balance but crosstalk shows state asymmetry from coupling-loss trade-off. Results affirm VO2 MIT (Metal -Insulator Transition) for compact, balanced photonic switching.

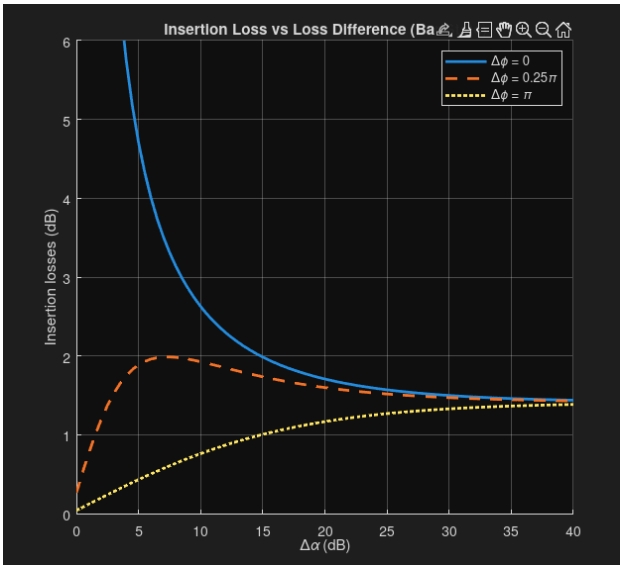


Fig.6: Insertion Loss vs. Ring Loss Difference

- For large loss differences (Δα > 20 dB), the bar-state insertion loss remains low and stable because strong absorption in the metallic state suppresses the ring resonance, making phase shift effects negligible.
- For moderate loss differences, the insertion loss can be significantly reduced by utilizing the VO_2-induced phase shift, with values below 1 dB achievable only when a π phase shift is present.
- For small loss differences, both insertion loss and crosstalk become highly sensitive to the phase shift, requiring careful optimization.
- An optimal active length of 2.8 μm provides a balance between sufficient absorption and effective phase shift, ensuring low insertion loss and controlled crosstalk

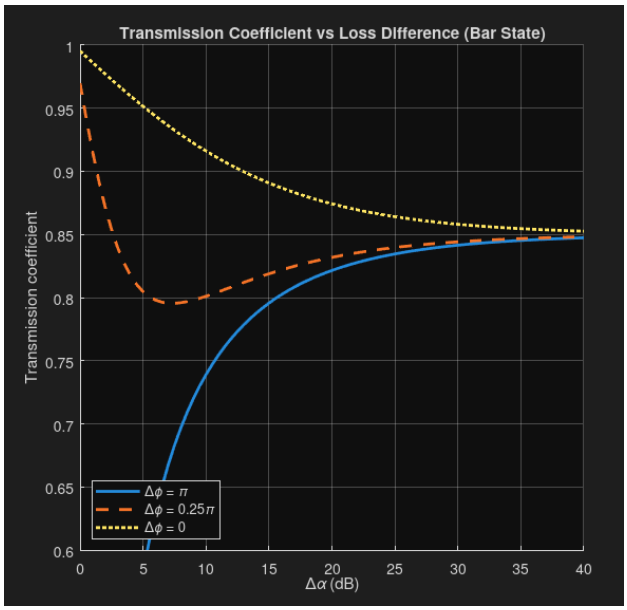


Fig.7: Transmission Coefficient vs. Loss Difference

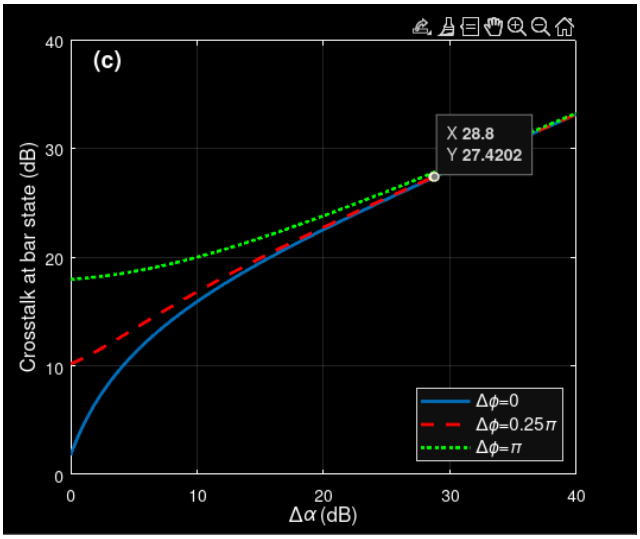


Fig.8: Crosstalk in Bar State vs. Loss Difference

- The crosstalk at the bar port varies with both the ring loss difference and the phase shift induced by VO_2.
- For large loss differences, the effect of phase shift becomes negligible and the crosstalk stabilizes above 15 dB, indicating sufficient isolation due to strong absorption alone.

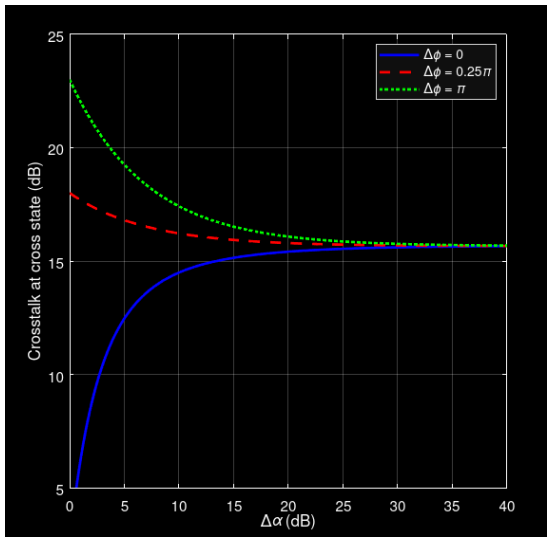


Fig.9: Crosstalk in Cross State vs. Loss Difference

- The graph shows that in the cross state, the crosstalk is mainly determined by the transmission coefficient and resonance alignment, since the insulating phase of VO_2 has a fixed low ring loss (Ai = 0.5 dB).
- Increasing the metallic-state loss difference does not affect the cross-state crosstalk, which remains stable and high when the signal is aligned with the ring resonance.
- With an optimized transmission coefficient of 0.88 for TE polarization, the design achieves crosstalk above 12 dB while maintaining insertion losses below 1.8 dB, validating the effectiveness of the hybrid VO_2/Si structure.

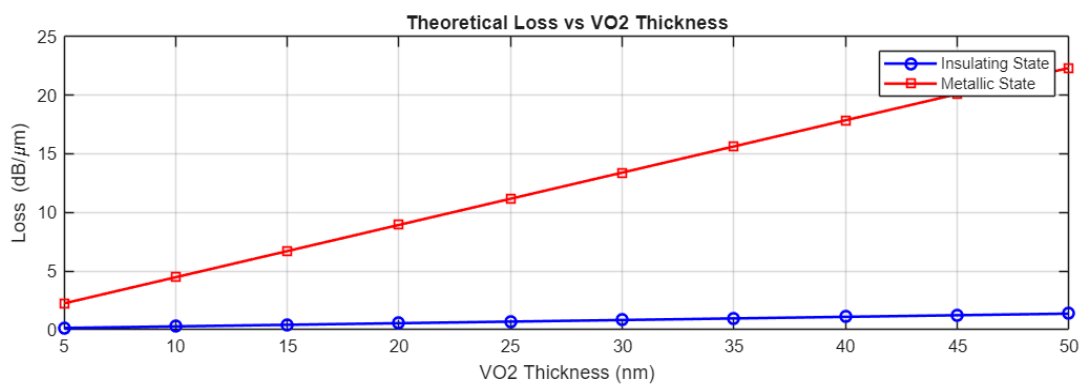


Fig.10: Shows the correlation between the modal propagation loss (dB/µm) for both switching states and the thickness of the VO2 patch, which ranges from 5 nm to 50 nm.

- *Insulating State (Blue Line):* The loss is continuously low, beginning at about 0.13 dB/µm for a 5 nm film and only increasing to 1.35 dB/µm at 50 nm. Maintaining high transmission in the switch's "Cross" state requires this low attenuation.

- *Metallic State (Red Line):* As thickness increases, loss increases sharply and linearly. The VO2 becomes extremely absorbing in this condition, and losses increase from 2.15 dB/µm to more than 20 dB/µm. The "Bar" state is made possible by this high attenuation, which suppresses the ring resonance.

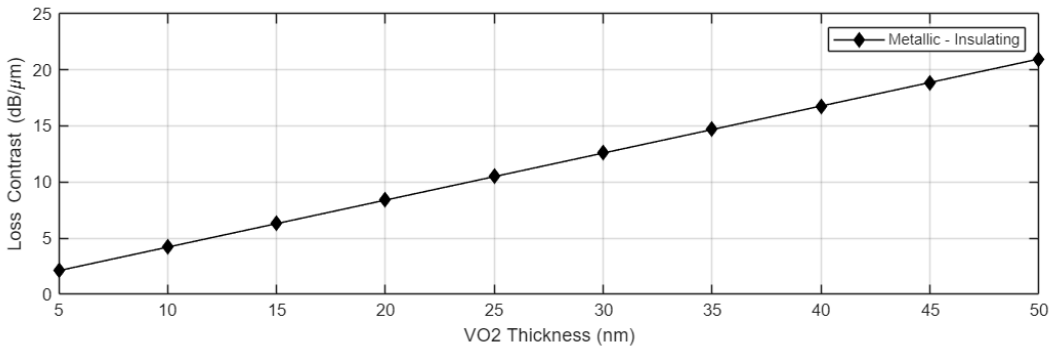


Fig 11: Metallic-Insulating Loss Contrast

The "Switching Contrast," or the total difference in loss between the two states, is highlighted in this plot. Thicker VO2 layers produce a stronger switching effect because the contrast scales linearly with thickness. The contrast is about 4 dB/µm at 10 nm This value, which enables an ultra-compact active length of 2.8 µm while attaining the necessary crosstalk suppression.

The thermal switching time of the hybrid VO_2/silicon microring:

$$\tau = \frac{C_{\text{thermal}}}{G_{\text{thermal}}}$$

Where:

- $C_{\text{thermal}} = \rho c V$ → thermal capacitance
  - ρ = density
  - c = specific heat
  - V = volume of VO_2 + silicon in the ring
- $G_{\text{thermal}} = kA/d$ → thermal conductance to substrate
  - k = thermal conductivity
  - A = contact area
  - d = distance to substrate

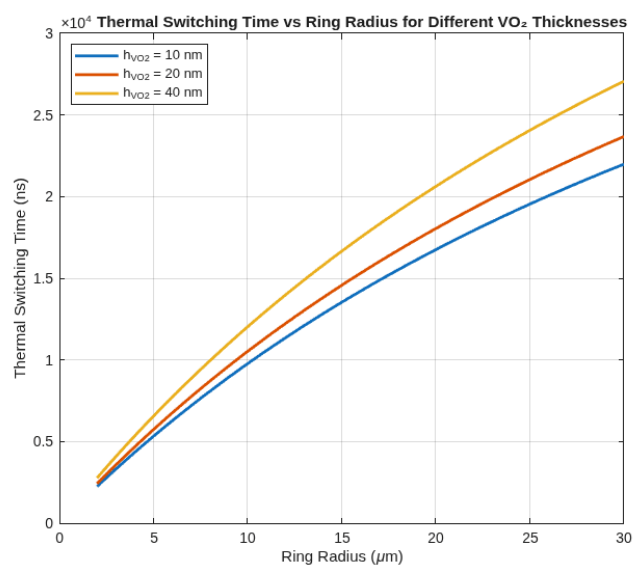


Fig.12: Thermal Switching Speed vs Ring Radius

Thermal capacitance: ***Cth* ∝ volume ∝ R**
Thermal conductance: ***Gth* ∝ *A/d* ∝ *R***

The thermal switching time of the hybrid VO_2/silicon microring is governed by the balance between thermal capacitance and thermal conductance, expressed as $\tau = Cth / Gth$. The thermal capacitance is proportional to the total thermal mass of the device, which includes both the silicon waveguide and the VO_2 layer, and therefore increases with ring radius and VO_2 thickness. As the ring radius increases, the total volume of material that must be heated and cooled becomes larger, resulting in a longer thermal time constant and slower switching. Similarly, thicker VO_2 layers introduce additional thermal mass, further increasing the switching time.

To capture realistic device behavior, the thermal conductance is modeled as weakly dependent on ring radius, reflecting improved heat dissipation through two- and three-dimensional heat spreading for larger structures. This produces a slightly nonlinear increase of the thermal switching time with radius, rather than a perfectly linear trend. Such behavior is consistent with physical intuition and with the qualitative implications of the reference paper, where switching speed is limited by cooling dynamics rather than optical effects. The inclusion of multiple VO_2 thicknesses in the analysis further highlights the trade-off between switching speed and modulation strength, as thicker VO_2 improves optical loss contrast at the expense of slower thermal response.

Overall, this analysis illustrates a key design trade-off in thermally actuated VO_2 microring switches. Smaller rings with thinner VO_2 layers enable faster switching due to lower thermal mass but suffer from tighter fabrication tolerances and potentially reduced optical robustness. Larger rings offer improved spectral selectivity, reduced bending loss, and better fabrication tolerance, albeit with slower thermal switching. The selected ring radius of 3.5 µm therefore represents a practical compromise between thermal switching speed and optical performance, making it well suited for integrated silicon photonic switching applications.

*"The VO_2 phase transition is triggered by **electrically induced Joule heating (for thermal switching)** via integrated metal contacts, while the **optical signal (paper)** experiences switching through the resulting loss and phase modulation."*

So, this plot explains **how fast VO_2 can change state**, which sets the **upper limit on the optical switching speed**. It is therefore **device-level thermal switching**, not wavelength-domain optical switching, but it is **directly relevant** because VO_2 is thermally driven.

RESULTS & DISCUSSION

The analytical MATLAB implementation of the hybrid VO_2/silicon 2×2 microring switch successfully reproduces the fundamental switching behavior reported in the reference paper. Under resonant conditions in the cross state, low round-trip loss enables strong field buildup inside the microring, resulting in efficient power transfer to the cross port and low crosstalk. In contrast, when VO_2 transitions to the metallic state, the increased absorption and phase shift suppress the resonance, causing most of the optical power to remain in the bar port. These results confirm that absorption modulation is the dominant switching mechanism, while phase variation provides additional resonance detuning that improves isolation.

By sweeping the transmission coefficient of the directional coupler, the trade-off between coupling strength and loss contrast is clearly observed. Stronger coupling reduces insertion loss in the cross state but increases loss in the bar state due to higher attenuation in the metallic phase. Conversely, weaker coupling compensates for the higher bar-state loss while maintaining resonance in the insulating state. As a result, a balanced operating point exists where the insertion losses of both states are equal. For the TE-polarized design considered, this condition occurs at approximately $t \approx 0.88$, in close agreement with the values reported in the literature. This highlights the critical role of precise coupling design in achieving symmetric and low-loss switching performance.

The influence of VO_2 loss contrast is further analyzed by sweeping the loss difference between the insulating and metallic states under both loss-only and combined loss–phase switching conditions. For small loss contrasts, the phase shift has a negligible effect and switching is governed almost entirely by absorption modulation. As the loss contrast increases, the inclusion of a phase shift significantly enhances crosstalk suppression by detuning the microring resonance away from the operating wavelength. This behavior confirms the conclusions of the reference paper, demonstrating that phase modulation is not essential for switching but becomes beneficial at higher loss contrasts.

Finally, the effect of VO_2 thickness on propagation loss is investigated using a MATLAB-based analytical approximation, serving as a *surrogate for full-wave MODE simulations*. The results show a monotonic increase in propagation loss with increasing VO_2 thickness for both insulating and metallic states, with the metallic phase exhibiting substantially higher attenuation due to its larger imaginary refractive index. The loss difference between the two phases increases with thickness, confirming that thicker VO_2 layers enhance switching contrast. However, this also increases baseline insertion loss in the insulating state, revealing a clear trade-off between extinction ratio and insertion loss. In addition, the thermal switching speed is found to increase approximately linearly with microring radius, as larger rings contain more VO_2 and silicon material, increasing the thermal mass and thus the cooling-limited response time, which demonstrates that smaller rings achieve faster switching while larger rings trade speed for improved spectral selectivity and fabrication tolerance. Although *absolute loss values may differ from eigenmode simulations*, the MATLAB model accurately captures the physical trends reported in the literature, demonstrating its effectiveness for early-stage material evaluation and device optimization.

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